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A Calibrated Spatio-Temporal Machine Learning Model for Tropical Peatland Fire Risk

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Abstract

Tropical peatland fires are complex environmental hazards with significant impacts. However, existing fire risk approaches, including empirical drought-fire indices and machine learning models that focus on either spatial or temporal aspects, remain limited in representing the dynamics of fire occurrence across heterogeneous peatland landscapes. Here, we develop an integrated spatio-temporal machine learning model with a probabilistic calibration method to analyze and predict tropical peatland fire risk. The framework employs a Convolutional Neural Network (CNN) to extract spatial features, Convolutional Long Short-Term Memory (ConvLSTM) to capture temporal dependencies, and isotonic regression to calibrate predicted fire probabilities. We employ meteorological, hydrological, land cover, anthropogenic, and historical fire data from tropical peatlands in Kalimantan, Indonesia. The proposed model achieves good performance and produces probability of fire maps that are consistent with observed fire hotspot distributions. Furthermore, the forecasting results show that the model can identify shifting fire risk patterns across peatland areas over different time horizons. These findings demonstrate that the proposed framework provides not only a predictive tool but also actionable information for early warning, risk assessment, and fire mitigation in tropical peatland ecosystems.

Keywords: Tropical peatland, Fire risk, Spatio-temporal model, Probability of fire, Machine learning model

1. Introduction

Tropical peatland fires are significant environmental hazards influenced by complex processes that evolve simultaneously across space and time. Analyzing and forecasting tropical peatland fire risk, including the location and timing of fire occurrence, can support risk assessment and fire mitigation efforts. This paper develops an integrated spatio-temporal machine learning model with a probabilistic calibration method to analyze and predict tropical peatland fire risk. The proposed framework combines three complementary methods. Convolutional Neural Network (CNN) is used to learn spatial patterns and identify areas with higher fire susceptibility. Convolutional Long Short-Term Memory (ConvLSTM) is applied to capture temporal dependencies and represent the evolution of fire prone conditions over time. Finally, isotonic regression is employed to calibrate the predicted fire probabilities, improving the consistency between model outputs and observed fire occurrence frequencies. Through this approach, the proposed model provides calibrated probability maps for examining and forecasting tropical peatland fire risk that are in agreement with observed fire hotspot distributions.

Tropical peatlands are critical ecosystems that contribute significantly to carbon storage, hydrological regulation, and the support of biodiversity (Alexandrov et al., 2020; Ledger et al., 2024; Loisel et al., 2021). Saturated conditions in tropical peatlands, associated with low hydraulic gradients (Baird et al., 2017), promote high rates of organic matter accumulation, resulting in the development of carbon rich ecosystems and contributing significantly to the global carbon cycle (Page et al., 2022). The high carbon storage capacity of undisturbed peat ecosystems limits greenhouse gas emissions, thereby contributing to climate change mitigation. By maintaining hydrological balance, tropical peatlands regulate water dynamics and reduce the intensity of hydrological extremes. The high water retention of tropical peatlands allows excess rainfall to be retained during wet periods and gradually

released during dry sea sons, which mitigates floods and maintains water availability (Lupascu et al., 2020; Mukhaiyar et al., 2024; Wu et al., 2026). Furthermore, the unique environmental conditions of tropical peatlands support habitats for biodiversity of high conservation value (Girkin et al., 2022). A primary example is the endangered orangutan, Asia's only remaining great ape, whose survival heavily depends among others on peat swamp forests (Felton et al., 2003).

Although tropical peatlands provide important functions to the environment, they are susceptible to severe and persistent fires that result in extensive impacts across multiple interconnected sectors. Recurrent fires degrade peat structure, reduce biodiversity, and disrupt hydrological processes that are critical to ecosystem resilience (Horton et al., 2022; Mezbahuddin et al., 2023; Wang et al., 2022). Large scale fire events release massive amounts of greenhouse gases and fine particulate matter into the atmosphere, contributing to climate change and severe air pollution (Rodríguez Vásquez et al., 2021). The resulting haze is associated with an increased incidence of respiratory and cardiovascular diseases across affected regions (Koplit et al., 2016; Kuwata, 2024), and contributes to both infant and adult mortality (Hein et al., 2022). Moreover, tropical peatland fires produce significant financial losses due to agricultural damage, infrastructure destruction, and supply chain interruption (Kiely et al., 2021; Situngkir et al., 2025). The estimated economic loss in Indonesia resulting from the 2015 tropical peatland fires was approximately US\$16 billion, equivalent to 1.8% of the country's GDP (Horton et al., 2021).

To support fire prevention strategies, drought-fire index models for tropical peatland have been developed (Mashuri et al., 2026; Taufik et al., 2022, 2015). For example, Taufik et al. (2022) proposed Peat Fire Vulnerability Index (PFVI), which incorporates hydrological and meteorological variables, including water table, soil moisture, rainfall, and air temperature. The PFVI provides interpretable metrics for assessing fire risk and classifies fire danger into four

categories. However, it remains limited by threshold-based structures, empirical parameters, and partial representation of nonlinear interactions among variables. Consequently, the PFVI may not effectively represent the complex feedback that govern fire occurrence across heterogeneous tropical peatland landscapes.

To address these limitations, machine learning approaches are employed to model complex interactions and improve the predictive accuracy of fire occurrence. Machine learning methods provides flexibility in capturing nonlinear relationships and interactions among multiple drivers without relying on predefined functional forms (Fauziah et al., 2025; Guria et al., 2024, 2025; Horton et al., 2021; Kurniawan et al., 2025; Mahdiyasa et al., 2025; Mezbahuddin et al., 2023; M. Mishra et al., 2024, 2025; Nie et al., 2026). However, the existing machine learning models of tropical peatland fire risk focus on either temporal or spatial aspects separately. For example, Mahdiyasa et al. (2025) developed a model *peatfr* to predict and analyze tropical peatland fire through Long Short-Term Memory and Gate Recurrent Unit machine learning methods. This model focuses primarily on temporal patterns and captures time dependent relationships associated with fire occurrence. However, *peatfr* provides limited information about the spatial distribution of fire risk. In contrast, Horton et al. (2021) proposed a fire risk model for tropical peatland using machine learning methods XGBoost and Random Forest. Although this approach successfully captured the key factors contributing to fire susceptibility, the model focuses mainly on spatial characteristics and does not explicitly account for temporal dynamics. Furthermore, the probability map produced by this model is not properly calibrated and may overestimate the likelihood of fire occurrence, which reduces the reliability of the model.

In this paper, we develop a novel spatio-temporal machine learning model with a probabilistic calibration method to analyze and forecast tropical peatland fire risk. The proposed model integrates Convolutional Neural Networks (CNN) to extract spatial features,

Convolutional Long Short-Term Memory (ConvLSTM) to learn temporal dependencies, and isotonic regression to calibrate the predicted probabilities of fire occurrence. Meteorological, hydrological, land cover, and anthropogenic variables are incorporated to capture the complex interactions that influence fire occurrence. This integrated approach aims to represent the processes governing fire ignition and propagation properly, which in turn support effective early warning systems and fire prevention strategies.

2. Methods

2.1 Study area

The study area is located in Kalimantan, Indonesia, which contains one of the largest tropical peatland ecosystems in Southeast Asia (**Figure 1**). Peatlands in this region cover approximately 4.54 million hectares, representing about 33.8% of total Indonesian peatlands (Anda et al., 2021). These peatlands are distributed in all five provinces, with the largest extents found in Central and West Kalimantan provinces. Peat depth in Kalimantan varies considerably, ranging from shallow deposits (around 50 cm) to deep peat layers (more than 700 cm) (Anda et al., 2021). Shallow peat deposits are generally found near peatland margins or transitional zones, while deeper peat layers are associated with central peat dome formations where long-term organic accumulation has occurred. The peatland distribution data employed in this study were obtained from the Geospatial Information Agency of Indonesia (BIG) (Geospatial Information Agency, 2019).

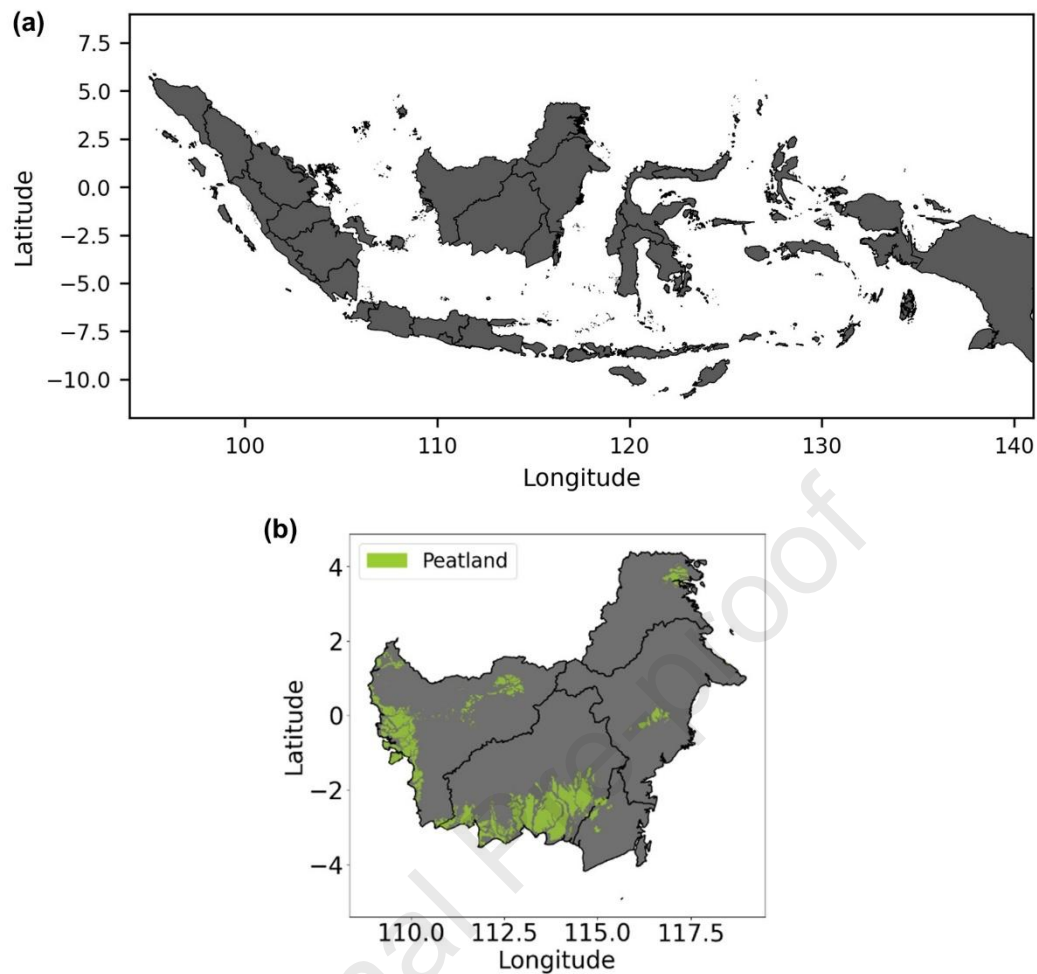


Figure 1. Location of the study area. (a) Map of Indonesia showing the geographic position of Kalimantan Island and (b) the distribution of tropical peatlands in Kalimantan, Indonesia.

Peatlands in Kalimantan have experienced increasing environmental pressure due to land-use change, drainage, and agricultural expansion (Horton et al., 2021, 2022). Consequently, the region frequently experiences peatland fires, particularly during dry seasons and prolonged drought periods associated with climate variability. Due to the large extent of peatlands and the high occurrence of fire events, Kalimantan provides a suitable and representative study area for investigating tropical peatland fire risk and developing spatio-temporal modeling approaches for fire prediction.

2.2. Data

2.2.1 Data Sources

Fire occurrence in this study was represented by hotspot detections obtained from the Fire Information for Resource Management System (FIRMS). The data were obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) Collection 6 (Giglio et al., 2016). For the analysis, only hotspot detections with a confidence level greater than 80% were included. The FIRMS dataset provides satellite-based observations of active fires and is used to monitor the spatial and temporal distribution of fire activity. Hotspot data were collected from 1 January 2020 to 31 December 2024 at daily temporal resolution (**Figure 2**).

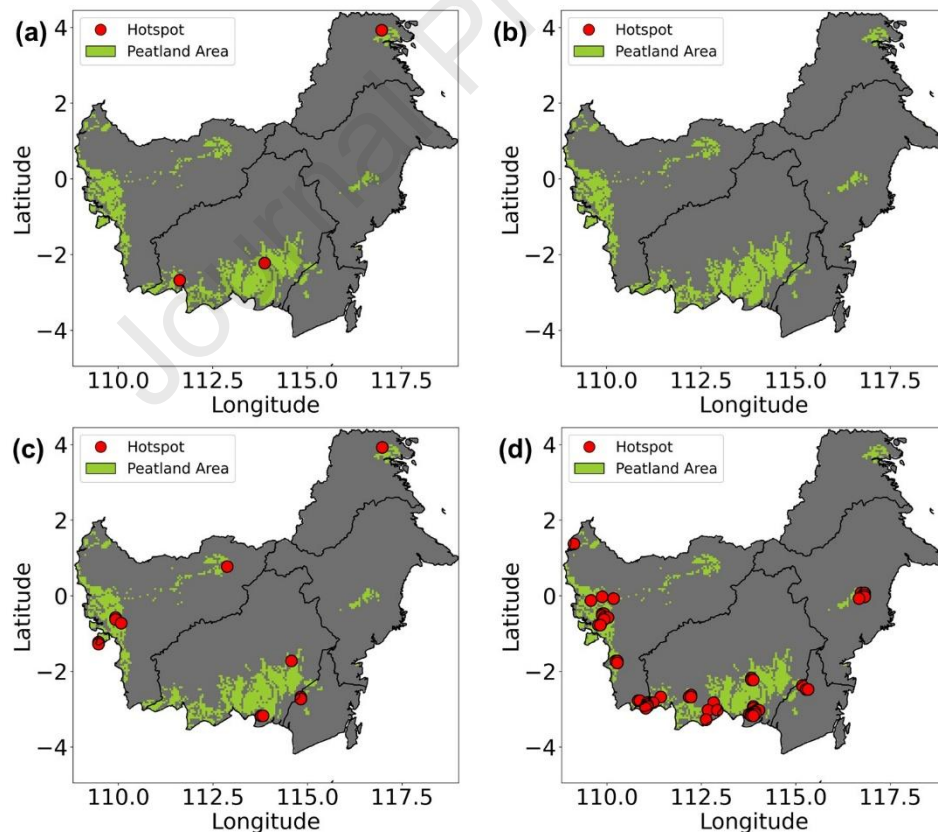


Figure 2. Fire hotspot data snapshots in the study area on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

Meteorological and hydrological conditions were also included to characterize factors influencing tropical peatland fire occurrence. These variables were collected over the same five-year period at daily resolution. Meteorological conditions were represented by rainfall (**Figure 3**) and air temperature (**Figure 4**), which regulate evapotranspiration and influence water table dynamics in peatland ecosystems. Rainfall data were obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (Funk et al., 2015), while air temperature was derived from the ERA5-Land reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) (Muñoz-Sabater et al., 2021). Hydrological conditions were represented by soil moisture (**Figure 5**), also obtained from ERA5-Land (Muñoz-Sabater et al., 2021). Soil moisture reflects the water content of the surface peat layer and directly affects fuel availability. Lower soil moisture reduces peat water content, increasing oxygen availability, and facilitating fire ignition and spread. Contrastingly, higher soil moisture maintains saturated conditions that limit oxygen diffusion and reduce the likelihood of sustained combustion.

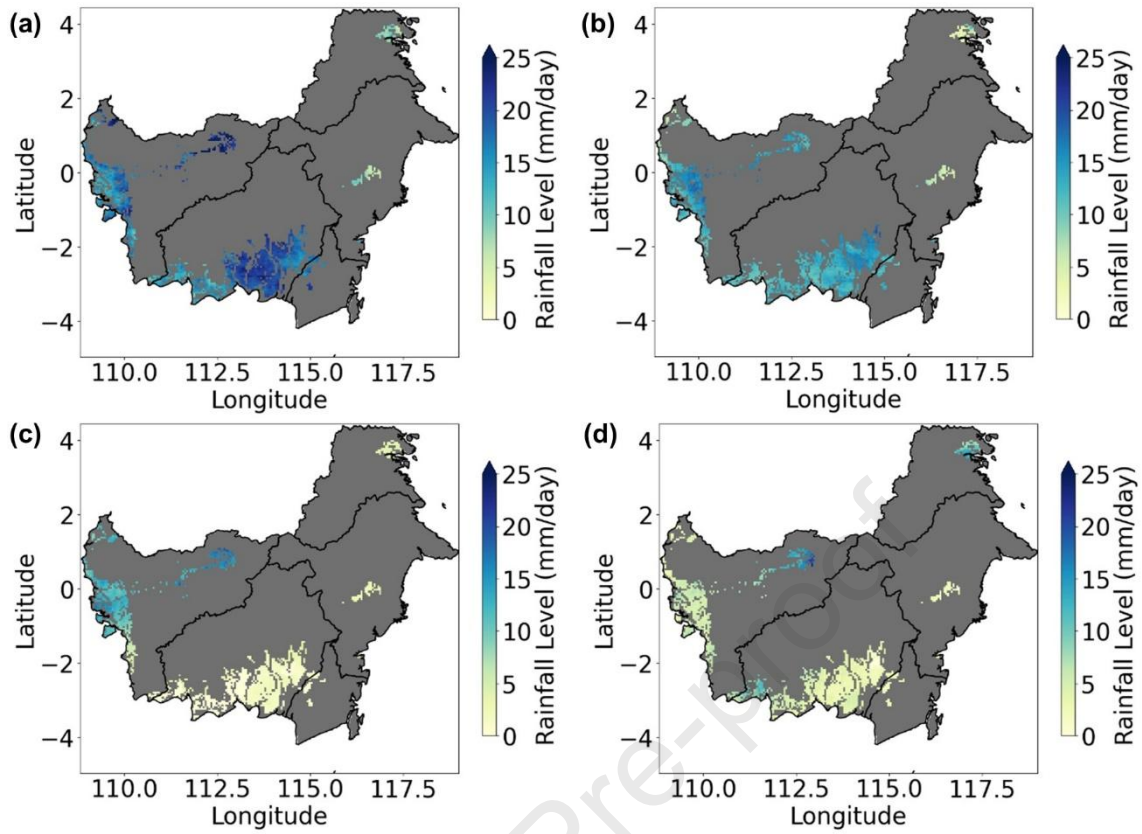


Figure 3. Rainfall data snapshots in the study area on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

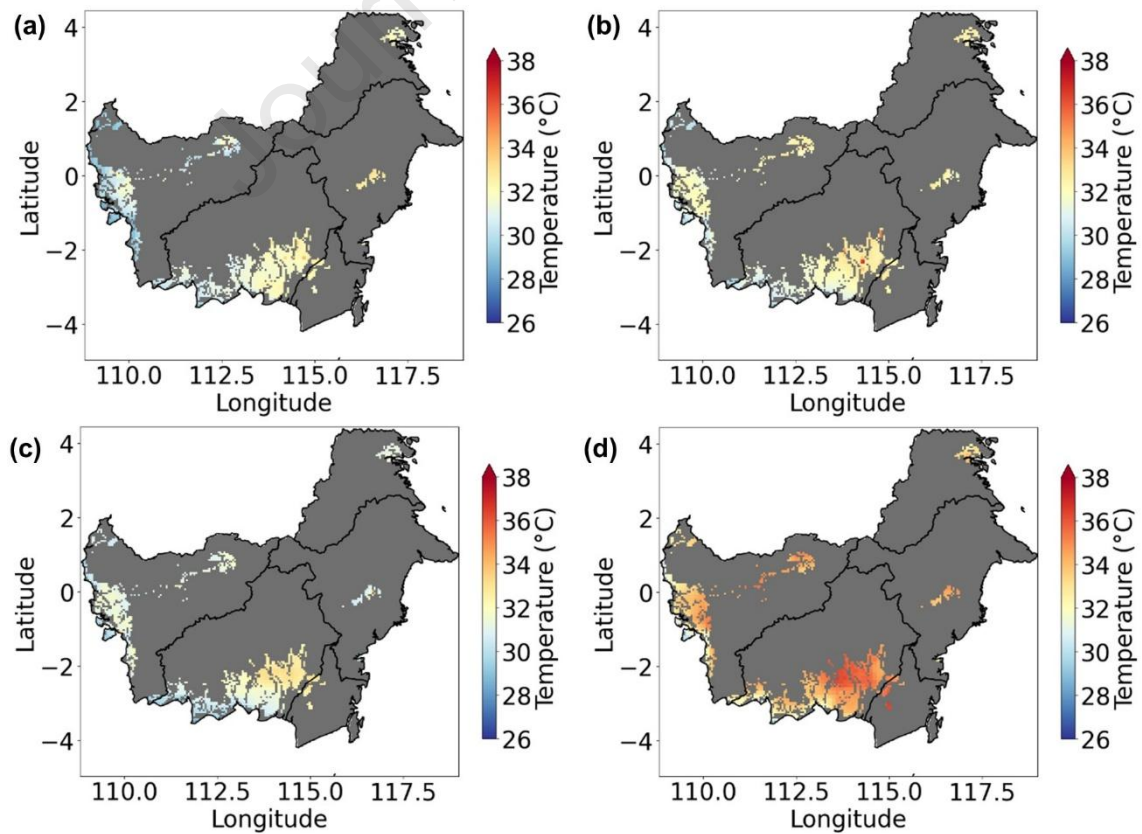


Figure 4. Temperature data snapshots in the study area on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

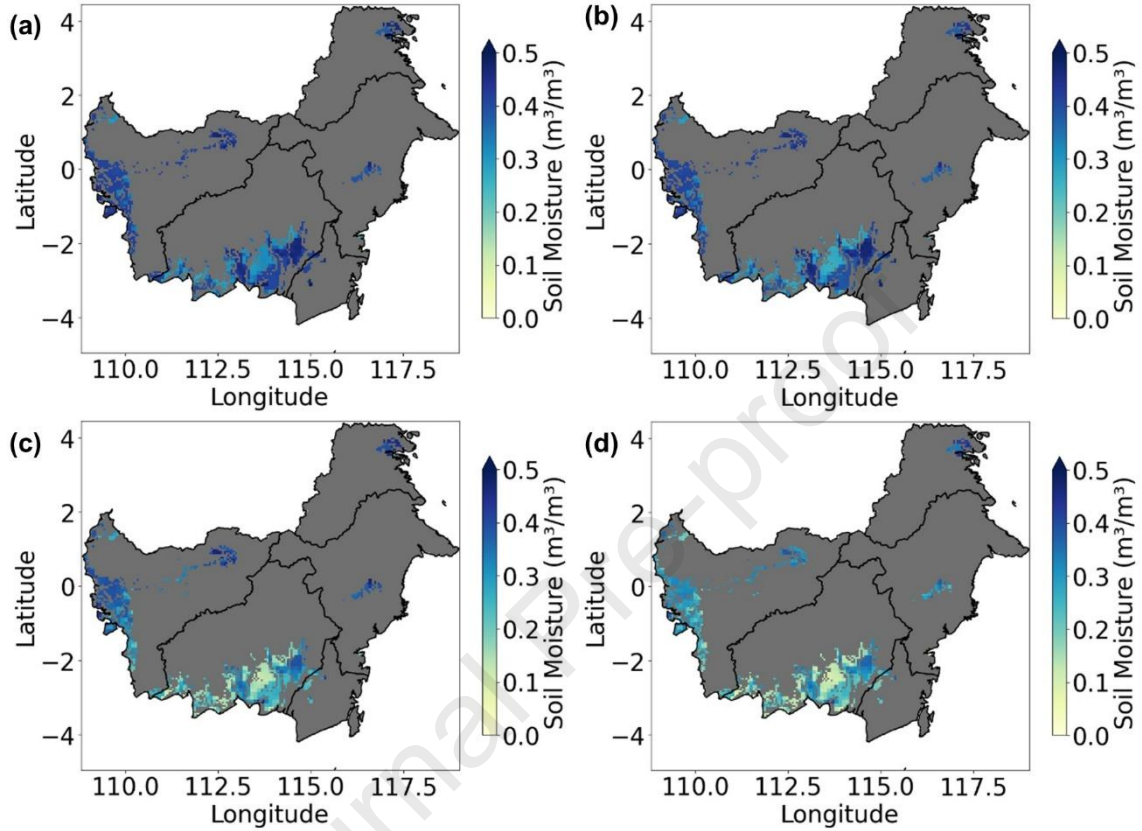


Figure 5. Soil moisture data snapshots in the study area on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

In addition to meteorological and hydrological variables, the dataset also incorporates land cover and distance to human activities (**Figure 6**) to represent the spatial characteristics of the landscape and human influences on fire occurrence. Land cover data were obtained from the MODIS Collection 6 Land Cover Type Product (Sulla-Menashe et al., 2019). This dataset characterizes the dominant vegetation or land use within each spatial unit, which determines the structure of available fuels, biomass accumulation, and baseline flammability conditions. Distance to human activities was derived from the Global Human Settlement Layer (GHSL)

developed by the European Commission Joint Research Centre (Pesaresi et al., 2024). This variable represents the proximity of each location to areas associated with human presence and is used as a proxy for anthropogenic influence on fire ignition. In tropical peatland regions, fires are often associated with land clearing, agricultural expansion, and other land management practices. As a consequence, proximity to human activities is an important variable influencing fire occurrence. **Table 1** summarizes the data and corresponding sources employed in this study, which provide the foundation for constructing the input variables for model development.

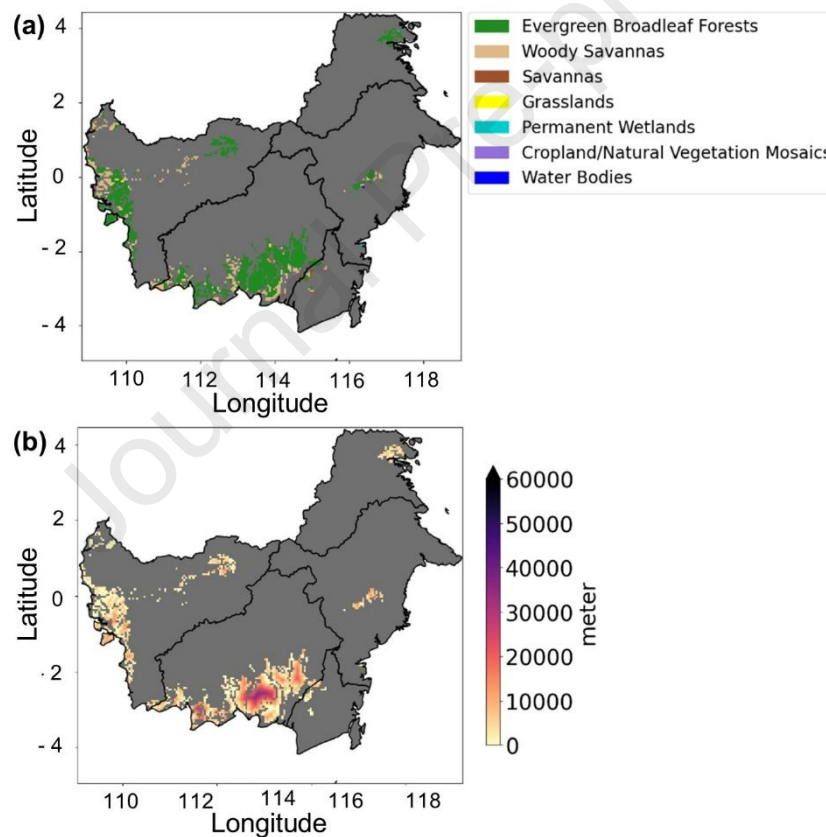


Figure 6. Spatial distribution of (a) land cover and (b) distance to human activities. Land cover indicates vegetation type and potential fuel availability across peatland areas. Distance to human activities represents anthropogenic pressure and potential fire ignition sources.

Table 1. Summary of datasets and corresponding sources employed in this study.

Data	Source	Reference
Peatland distribution	Geospatial Information Agency of Indonesia (BIG)	Geospatial Information Agency (2019)
Hotspot	Fire Information for Resource Management System (FIRMS)	Giglio et al. (2016)
Rainfall	Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)	Funk et al. (2015)
Air temperature	ERA5-Land reanalysis dataset	Muñoz-Sabater et al.(2021)
Soil moisture	ERA5-Land reanalysis dataset	Muñoz-Sabater et al.(2021)
Land cover classification	MODIS Collection 6 Land Cover Type Product	Sulla-Menashe et al.(2019)
Distance to human activity	Global Human Settlement Layer (GHSL)	Pesaresi et al.(2024)

2.2.2 Data Preprocessing

Prior to modeling, the datasets were preprocessed to ensure spatial consistency and compatibility for machine learning analysis. Because the data were obtained from multiple sources with different spatial resolutions and formats, all variables were resampled to a uniform spatial grid with a resolution of 5×5 km. This resolution was selected to align all variables within the same spatial reference system and to minimize potential biases arising from

differences in spatial scale. The use of a 5×5 km grid is also consistent with previous grid based fire analyses. For example, Tan et al. (2020) resampled multi source fire related datasets to a 5 km grid in an analysis of forest and land fires in Indonesia, noting that this resolution represented a trade-off between spatial accuracy and computational feasibility. Similar 5×5 km grid structures have been used to summarize fire counts and model fire occurrence in the Western Himalayas (Bar et al., 2023), and to analyze hotspot density and dry lightning wildfire probability in Tasmania (Wickramasinghe et al., 2024). However, the 5×5 km grid should be interpreted as a regional scale fire occurrence probability unit rather than a fine scale fire patch detection resolution.

To quantify the spatial dependence of the continuous input variables, global Moran's I was calculated after all datasets were resampled to the 5×5 km spatial grid. The analysis was applied to rainfall, air temperature, soil moisture, and distance to human activities, which are represented as continuous gridded predictors. For the time-varying variables, including rainfall, air temperature, and soil moisture, Moran's I was calculated from daily spatial maps, and the annual average Moran's I and percentage of statistically significant daily maps were summarized. The results show consistently positive spatial autocorrelation across all variables. Rainfall showed the highest spatial dependence, with annual average Moran's I values ranging from 0.910 to 0.934. Air temperature also showed strong spatial autocorrelation, with annual average Moran's I values ranging from 0.815 to 0.875. Soil moisture showed positive spatial autocorrelation, with annual average Moran's I values ranging from 0.742 to 0.780. In all years, 100% of the daily maps for these three variables were statistically significant. Distance to human activities also exhibited strong positive spatial autocorrelation, with a Moran's I value of 0.800. These results indicate that the continuous input layers contain strong spatially structured information, supporting the use of CNN-based convolutional filters to extract local spatial features from neighboring grid cells.

Before normalization, the stationarity of the time-varying continuous input variables was evaluated using the Augmented Dickey-Fuller (ADF) test. The test was applied to rainfall, air temperature, and soil moisture to assess whether these variables contained a unit root. The null hypothesis of the ADF test is that the time series is non-stationary. The results showed that the null hypothesis was rejected for all three variables at the 5% significance level. Rainfall had an ADF statistic of -4.5865 with a p-value of 0.0001, air temperature had an ADF statistic of -4.0835 with a p-value of 0.0010, and soil moisture had an ADF statistic of -4.2116 with a p-value of 0.0006. These results indicate that rainfall, air temperature, and soil moisture were stationary before normalization.

Continuous variables, including rainfall, air temperature, soil moisture, and distance to human activity, were normalized to the range [0, 1] to ensure comparable numerical scales and prevent bias toward features with larger magnitudes during model training (Horton et al., 2021). The normalization is expressed as

$$x_i^* = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where x_i is the original value, $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the corresponding variable, and x_i^* is the transformed value after normalization. Land cover data, originally represented as categorical integer labels, were transformed using one-hot encoding to ensure that each class was treated as an independent nominal category rather than an ordinal value. For the target variable, a grid cell was labeled as a fire if at least one hotspot detection was recorded within the cell. Conversely, grid cells without any detected fire events were labeled as non-fire. This binary classification approach enables the model to learn spatial and temporal patterns associated with fire occurrence across the study area.

To address the challenge of class imbalance between fire and non-fire classes, a weighted loss function was employed (Y. Ho & S. Wookey, 2020) during data preprocessing

$$L_w = - \sum_{i=1}^n (w y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)) \quad (2)$$

where L_w is the weighted loss function, n is the number of samples, y_i is the label for i^{th} sample, $\hat{y}_i$ is the predicted probability of fire occurrence produced by the model, w is the positive constant weight coefficient. Non-fire observations typically dominate the dataset which can bias the model toward predicting the majority class. The weighted loss function assigns higher penalty weights to the minority class, enabling the model to better learn patterns associated with fire occurrences and improving its sensitivity to rare fire events.

2.3 Model

The proposed modeling framework is designed to capture the complex spatio-temporal dynamics of tropical peatland fire occurrence by integrating machine learning architectures with a probabilistic calibration method. The framework combines Convolutional Neural Networks (CNN) for spatial feature extraction, Convolutional Long Short-Term Memory (ConvLSTM) for learning temporal dependencies, and isotonic regression for probability calibration. This integrated approach enables the model to represent the spatio-temporal evolution of fire risk and produce calibrated probabilistic predictions suitable for risk analysis (Figure 7).

2.3.1 Model formulation

CNN is employed to extract spatial features from tropical peatland fire predictor maps at each observation time (Khirani et al., 2026; Yar et al., 2023; Zelioli et al., 2025). The main operation in CNN is the convolution process, which is defined as

$$F_t(i, j) = \sum_m \sum_n X_i(i - m, j - n) K(m, n) \quad (3)$$

286 where $F_t(i, j)$ is the resulting feature value at spatial position (i, j) , X_t is the input of spatial
 287 grid, and K is the convolution kernel. After convolution process, a nonlinear activation function
 288 is introduced to enhance the model capacity through the Rectified Linear Unit (ReLU)

$$\text{ReLU}(x) = \max(0, x) \quad (4)$$

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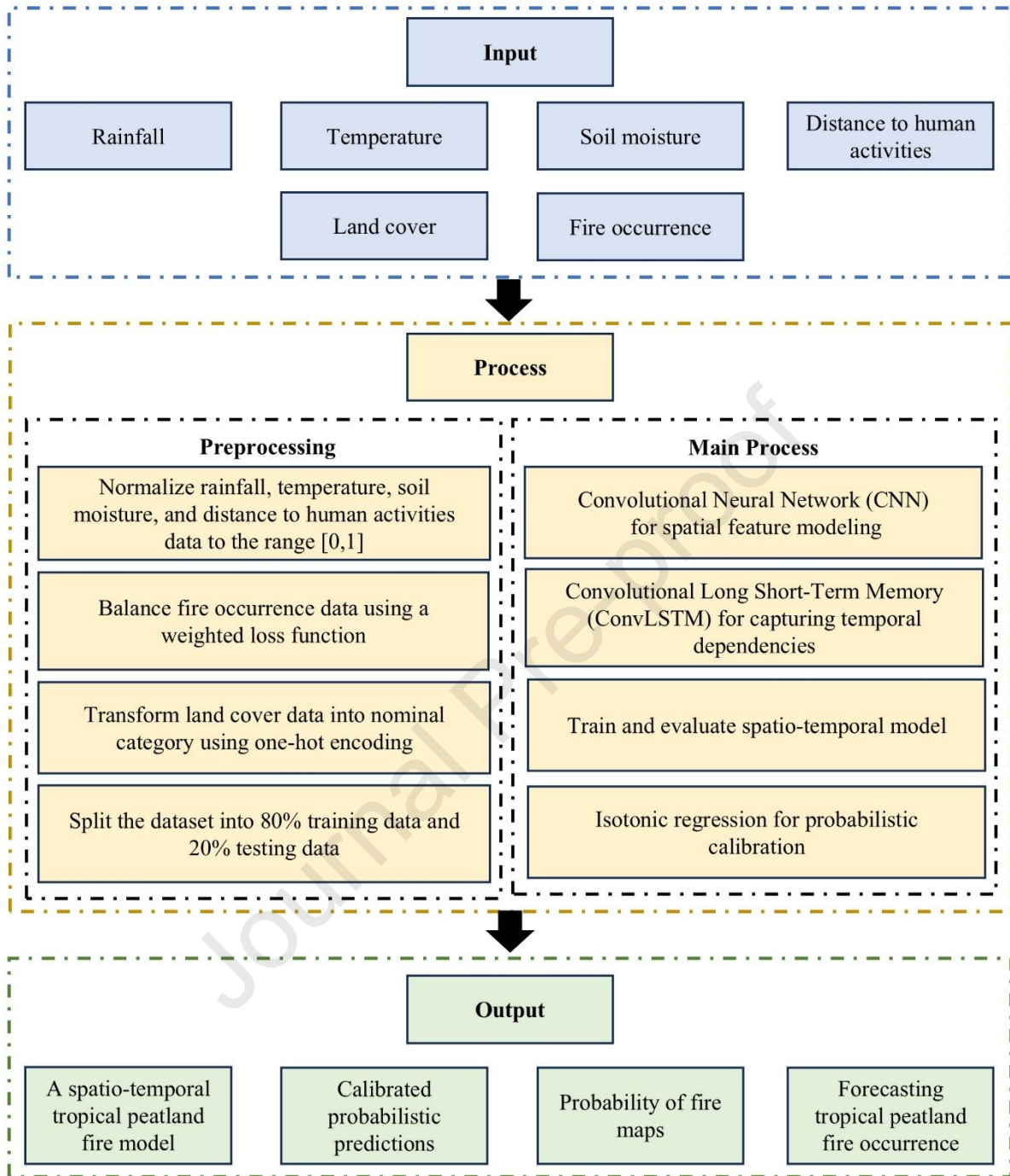


Figure 7. Conceptual diagram of the proposed model consisting of input, process, and output stages. The input stage includes meteorological, hydrological, land cover, anthropogenic, and fire occurrence variables. The process stage involves preprocessing data and main modeling process. Preprocessing data is conducted through normalization, class balancing using a weighted loss function, one-hot encoding of land cover data, and

training–testing data partitioning. The main modeling process integrates CNN for spatial feature extraction, ConvLSTM for temporal dependency learning, and isotonic regression for probabilistic calibration. The output stage produces calibrated probability maps for analyzing and forecasting tropical peatland fire, which support early warning, risk assessment, and fire mitigation efforts

The spatial feature maps extracted from several consecutive time steps are arranged into a spatio-temporal sequence and passed to a ConvLSTM layer, which extends the standard LSTM architecture by replacing matrix multiplication with convolution (Khalaf et al., 2024; Mambile et al., 2025; Xu et al., 2025). Consequently, it can preserve spatial information and capture temporal dependencies. The ConvLSTM equations are written as

$$i_t = \sigma(W_{xi} * F_t + W_{hi} * H_{t-1} + b_i) \quad (5)$$

$$f_t = \sigma(W_{xf} * F_t + W_{hf} * H_{t-1} + b_f) \quad (6)$$

$$o_t = \sigma(W_{xo} * F_t + W_{ho} * H_{t-1} + b_o) \quad (7)$$

$$\tilde{C}_t = \tanh(W_{xc} * F_t + W_{hc} * H_{t-1} + b_c) \quad (8)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (9)$$

$$H_t = o_t \odot \tanh(C_t) \quad (10)$$

where σ sigmoid activation function, $*$ is the convolution operator, $\odot$ is the element-wise multiplication, i_t is the input gate, f_t is the forget gate, o_t is the output gate, $W_{xi}, W_{xf}, W_{xo}, W_{xc}, W_{hi}, W_{hf}, W_{ho}, W_{hc}$ are the trainable convolution weights, b_i, b_f, b_o, b_c are the bias terms, F_t is the spatial feature map, $\tilde{C}_t$ is the candidate cell state, C_t is the updated cell state, and H_t is the hidden state. The final hidden state is mapped to the predicted fire risk through an output convolution layer

$$\hat{R}_t = g(H_t) \quad (11)$$

where $\hat{R}_t$ denotes the predicted spatial distribution of tropical peatland fire risk and g is the output mapping function.

To improve the reliability of the predicted outputs, isotonic regression (Delong & V. Wüthrich, 2025; Gneiting & Resin, 2023) is applied as a post processing calibration step. Although the CNN–ConvLSTM model can effectively capture relative fire susceptibility, its raw outputs may not accurately correspond to the true probability of fire occurrence. Isotonic regression addresses this issue by learning a monotonic mapping from the raw prediction to a calibrated probability

$$p_k = m(\hat{R}_k) \quad (12)$$

with the monotonicity constraint

$$\hat{R}_k \leq \hat{R}_l \Rightarrow m(\hat{R}_k) \leq m(\hat{R}_l) \quad (13)$$

where p_k is the calibrated probability, m is the calibration function, and $\hat{R}_k$ is the raw prediction from the CNN–ConvLSTM model. The calibration function is estimated by minimizing the squared error between observed outcomes and calibrated values

$$\min_m \sum_{k=1}^N (Y_k - m(\hat{R}_k))^2 \quad (14)$$

where N is the number of data points in the calibration dataset and Y_k is the observed fire occurrence indicator. This calibration preserves the rank ordering of the original predictions and improves agreement between predicted values and observed event frequencies. Therefore, the proposed framework provides both spatio-temporal model of tropical peatland fire risk dynamics and calibrated probabilistic outputs suitable for risk assessment and early warning applications.

2.3.2 Model implementation

The prepared data were partitioned into training and testing sets using a chronological split to preserve temporal ordering. The dataset covered the period from 1 January 2020 to 31 December 2024. The first 80% of the observations, corresponding to 1 January 2020 to 31 December 2023, were used for model training, while the last 20%, corresponding to 1 January 2024 to 31 December 2024, were reserved for testing. This chronological partition ensures that the model is evaluated on unseen future data and reduces the risk of information leakage from later observations into the training process.

The CNN–ConvLSTM architecture consisted of four main layers. The first layer was a ConvLSTM2D layer employed as the spatio-temporal feature extractor, with 32 filters, a 3×3 kernel size, tanh activation, and sigmoid recurrent activation. Batch normalization and dropout with a rate of 0.30 were applied after this layer to stabilize training and reduce overfitting. The second layer was a Conv2D decoder layer for extracting deeper spatial features, with 64 filters, a 3×3 kernel size, and ReLU activation, followed by batch normalization and dropout with a rate of 0.30. The third layer was a Conv2D feature summary layer with 32 filters, a 3×3 kernel size, and ReLU activation, followed by batch normalization. The final layer was a Conv2D output layer with one filter and a 1×1 kernel size. A sigmoid activation function was used in the output layer to produce one probability map with values constrained between 0 and 1. The model was trained using a batch size of 2. During the initial hyperparameter search, the maximum number of epochs was set to 70, with early stopping applied using a patience of 20 epochs. For the deeper training search, early stopping was also applied, with a patience of 30 epochs. These early stopping criteria were used to prevent overfitting and to terminate training when the validation performance no longer improved.

Because fire events were relatively rare compared with non-fire observations, the training data exhibited substantial class imbalance. To address this issue, a positive class weight was applied during model training. Several candidate positive class weights, including 10, 25,

50, 75, and 100, were evaluated using the testing period to examine the precision–recall trade off. Based on this evaluation, a positive class weight of 75 was selected because it provided the most balanced detection performance, with precision of 0.699 and recall of 0.885 (**Table 2**). This setting was therefore used in the final model to improve sensitivity to fire events while maintaining acceptable precision

Table 2. Precision and recall values obtained under different positive class weights. The positive class weight of 75 was selected because it provided the best balance between precision and recall for detecting fire occurrence in the testing period.

Positive class weight	Precision	Recall
10	0.719	0.025
25	0.464	0.703
50	0.389	0.325
75	0.699	0.885
100	0.459	0.793

3. Results

3.1 Model performance

We assessed model performance by examining both discrimination ability and probability reliability. The Receiver Operating Characteristics (ROC) curve shows that the proposed model achieves an Area Under the Curve (AUC) of 0.845, which indicates good classification performance in distinguishing fire occurrence from non-fire conditions (**Figure 8a**). Model performance is further evaluated through calibration analysis. The raw probability outputs from the model are not properly calibrated, because the calibration curve falls below the ideal diagonal line with the value of Brier score is around 0.198. This pattern indicates a discrepancy between the predicted probabilities and the observed event frequencies. Following

isotonic regression, the calibrated curve shifts closer to the ideal relationship, particularly over the probability range where the majority of predictions are distributed with the value of the Brier score decreased substantially to approximately 0.011 (Figure 8b).

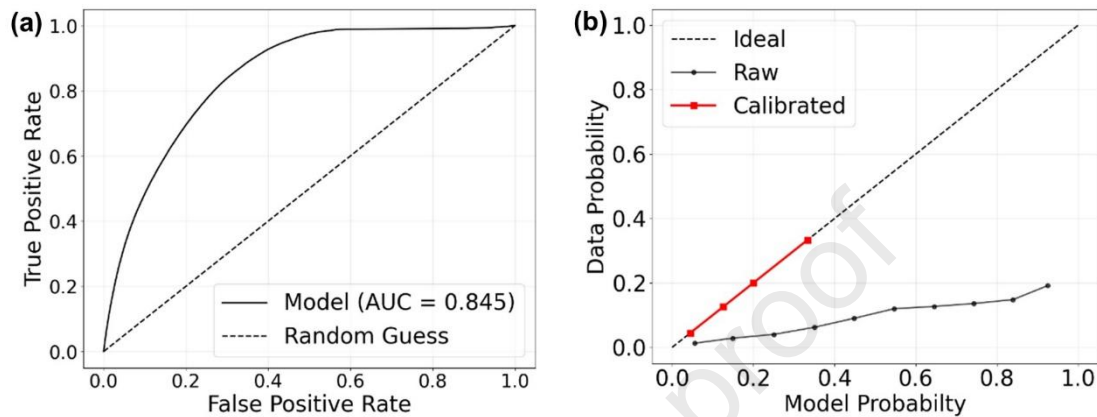


Figure 8. Model performance evaluation based on (a) the ROC–AUC curve and (b) the probability curve. The ROC–AUC curve shows the discriminatory ability of the model, while the probability curve examines the consistency between predicted probabilities and observed fire occurrence frequencies.

To evaluate whether model performance was robust under contrasting seasonal conditions, the testing period was divided into dry and wet seasons. The dry season was defined as May–October, while the wet season was defined as November–April. For each season, ROC–AUC was calculated to assess the discrimination ability of the model, while the Brier score was used to evaluate the reliability of the predicted probabilities.

The seasonal evaluation shows that the model maintained stable discrimination performance across both seasons. The ROC–AUC values were 0.811 for the dry season and 0.822 for the wet season, which are broadly consistent with the overall ROC–AUC of 0.845 obtained for the full testing period. This indicates that the model was able to distinguish fire and non-fire conditions under both fire prone dry conditions and wetter periods with lower fire

activity. The Brier scores for the dry and wet seasons were 0.088 and 0.027, respectively. These values are also consistent with the overall Brier score of around 0.011 for the full testing period. The higher Brier score during the dry season suggests greater probabilistic uncertainty under more fire prone conditions, whereas the lower wet season Brier score indicates relatively more reliable probability estimates during periods of reduced fire occurrence. These seasonal results support the robustness of the proposed model across contrasting hydroclimatic conditions.

The probability of fire results (**Figure 9**) indicate that the proposed spatio-temporal modeling framework is able to capture the main patterns of tropical peatland fire occurrence in Kalimantan. The spatial distribution of predicted fire probability indicates that fire risk is not uniformly distributed across the study area but is concentrated in peatland areas associated with woody savanna land cover and short distances to human activities. Furthermore, the model also captures the temporal variability of fire activity, with higher probabilities during the drier months of August and September, when peatland conditions become more vulnerable to ignition and fire spread.

To further evaluate the temporal robustness of the model, a rolling origin validation strategy was conducted in addition to the main chronological 80:20 train-test split (Farghaly et al., 2026; Fiorucci & Louzada, 2020). This validation strategy preserves the temporal structure of the data and avoids information leakage from future observations into the training period. Three rolling folds were evaluated. In Fold 1, data from 2020–2021 were used for training and data from 2022 were used for validation. In Fold 2, data from 2020–2022 were used for training and data from 2023 were used for validation. In Fold 3, data from 2020–2023 were used for training and data from 2024 were used for validation.

The rolling validation results show that the model generally maintained acceptable discrimination performance across different validation years. The ROC–AUC values were 0.8622 for Fold 1, 0.6987 for Fold 2, and 0.8538 for Fold 3. The minimum and maximum

ROC–AUC values across repeated runs were 0.8520–0.8784 for Fold 1, 0.6844–0.7211 for Fold 2, and 0.7211–0.8639 for Fold 3. The corresponding Brier scores were 0.0872, 0.1891, and 0.1729, respectively. The lower ROC–AUC and higher Brier score in Fold 2 indicate that the 2023 validation period was more difficult to predict, suggesting stronger interannual variability in fire risk conditions. The high ROC–AUC values in Folds 1 and 3 indicate that the proposed model can capture temporally transferable fire risk patterns. The rolling validation supports the robustness of the proposed framework while highlighting the need to account for interannual variability in operational fire forecasting.

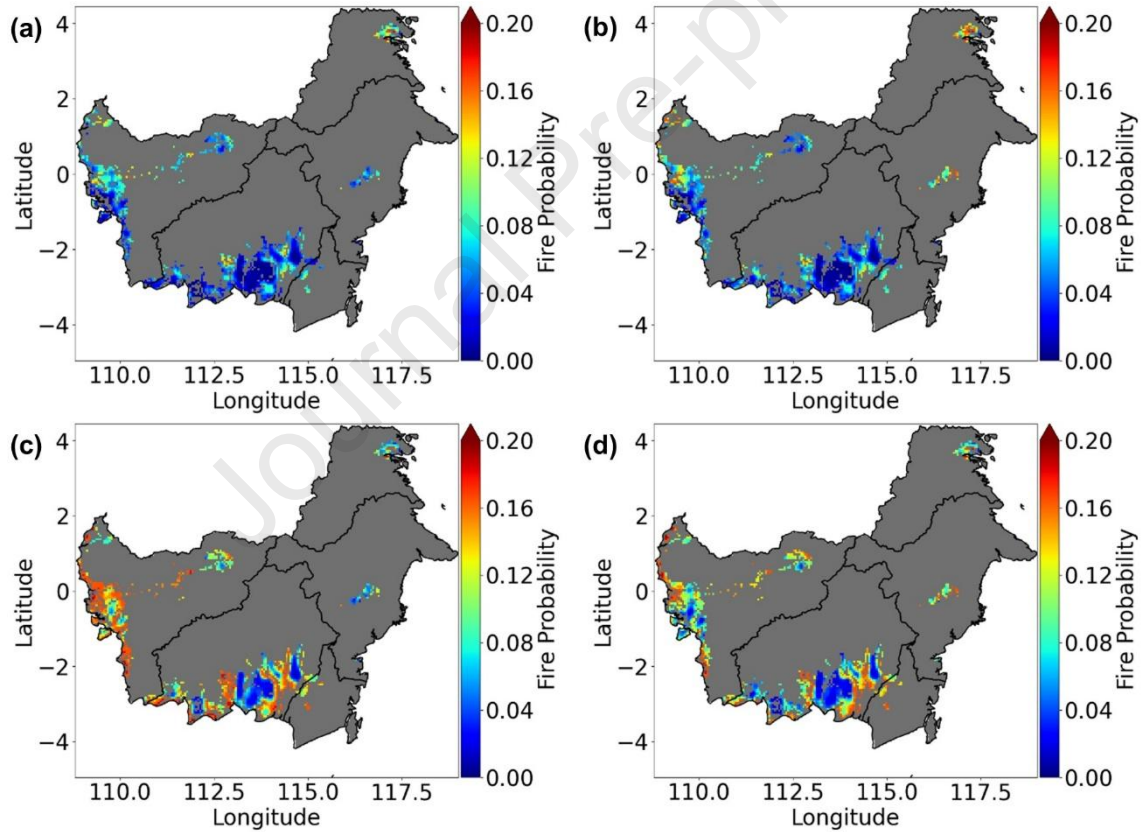


Figure 9. Probability maps of tropical peatland fire risk on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

The uncertainty analysis based on Monte Carlo dropout indicates that the model produced relatively narrow prediction intervals for areas with consistently low fire probability and wider intervals in transition zones between low and elevated fire risk (Figure 10). Across the testing period, the average width of the 95% prediction interval was 0.041 for low-risk grid cells, 0.083 for moderate-risk grid cells, and 0.112 for high-risk grid cells. This pattern suggests that prediction uncertainty increases in areas where environmental and hydrological conditions are closer to fire-prone thresholds.

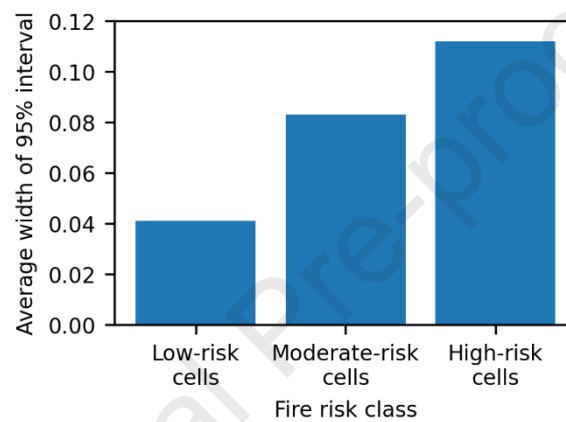


Figure 10. Monte Carlo dropout based uncertainty across fire risk classes. The average width of the 95% prediction interval increased from low-risk grid cells to high-risk grid cells, indicating greater predictive uncertainty in areas with elevated fire probability.

3.2 Forecasting capability

The forecasting capability of the proposed spatio-temporal modeling framework is evaluated by comparing the predicted probability of fire maps for the next three and ten days with the observed hotspots (Figure 11). In terms of spatial prediction, the forecast fire risk patterns are in agreement with the actual hotspot occurrences, because the observed hotspots are located within or adjacent to the elevated risk zones identified by the model. In the next three-day forecast, the observed hotspot was located in an area with a mean predicted fire probability of 0.284 and a 95% prediction interval of 0.221–0.336 (Figure 12). This indicates

that the area was consistently classified as having elevated fire probability across repeated stochastic predictions. In the next ten-day forecast, the observed hotspot was located in an area with a mean predicted fire probability of 0.247 and a 95% prediction interval of 0.176–0.319. Although the uncertainty interval was wider for the ten-day forecast, the hotspot remained within or adjacent to the elevated-risk zone. These results show that the model can identify fire prone areas while also quantifying the uncertainty associated with the forecasted probability maps.

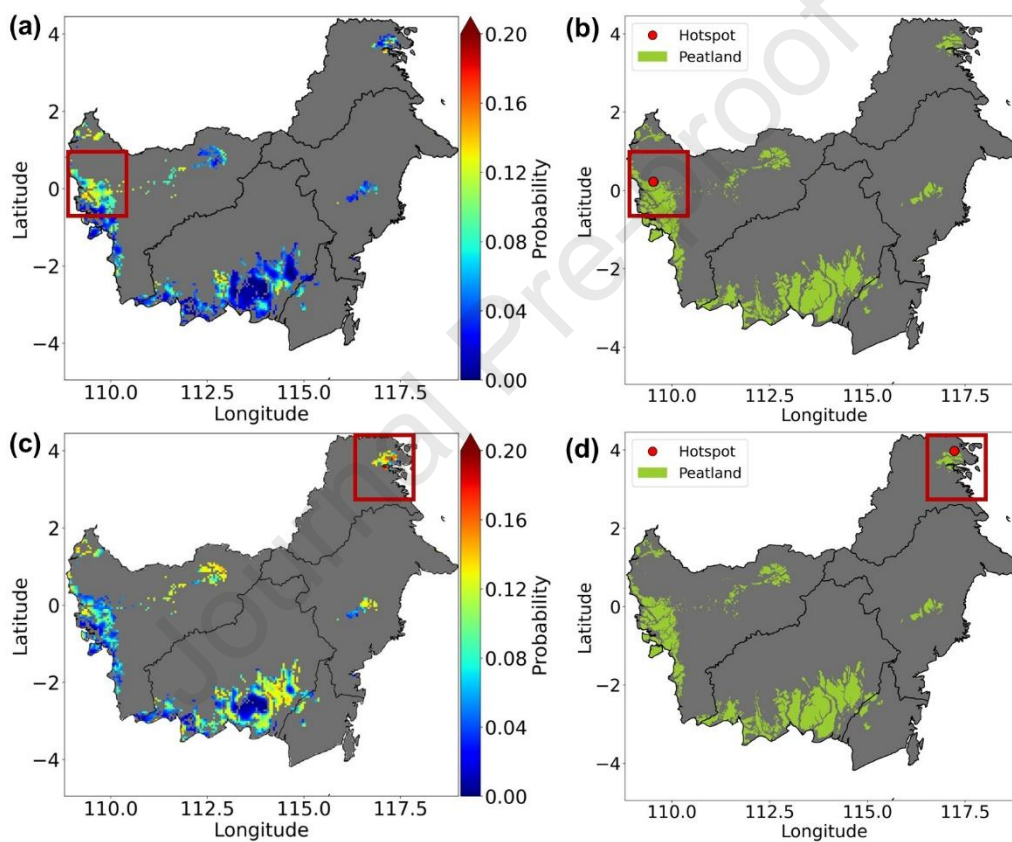


Figure 11. Comparison between forecasted fire probability from the proposed model and actual fire hotspots for (a, b) the next three-day forecast and (c, d) the next ten-day forecast. The red box highlights areas where high predicted fire probability aligns with observed fire hotspots.

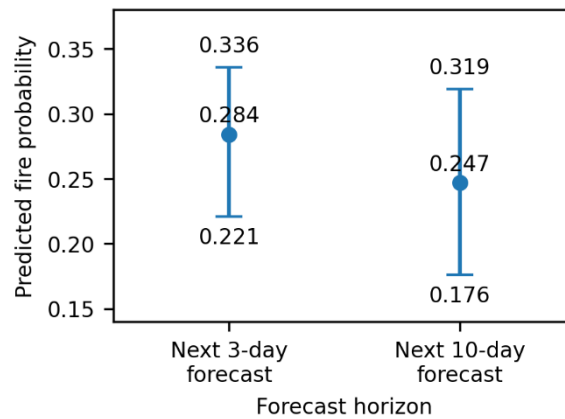


Figure 12. Forecasted fire probability with 95% prediction intervals for observed hotspot areas. The next ten-day forecast showed a wider prediction interval than the next three-day forecast, indicating greater uncertainty at the longer forecast horizon.

The forecasts also show that the fire risk distribution changes across the two prediction horizons. For the next three-day forecast, the model identifies an area of increased fire probability in the western peatland region, with the observed hotspot located within the predicted high risk area. For the next ten-day forecast, the predicted fire risk distribution shifts and emphasizes elevated probabilities in the northern region, where the observed hotspot is also recorded. These results suggest that the model is able to capture the evolution of fire prone conditions over time.

To complement the visual comparison between predicted fire probability maps and observed hotspots, quantitative validation was conducted for both forecast horizons. The evaluation was performed at the grid-cell level by comparing the predicted fire probabilities with the observed fire occurrence labels during the corresponding forecast periods. Three metrics were used: ROC–AUC, precision, and recall. The next three-day forecast achieved a ROC–AUC of 0.836, precision of 0.688, and recall of 0.842. These values indicate good short-term discrimination ability and show that most observed fire occurrences were captured within predicted elevated risk areas. For the next ten-day forecast, the ROC–AUC decreased to 0.784,

with precision of 0.670 and recall of 0.764. The lower performance for the ten-day horizon reflects the greater uncertainty associated with longer forecast periods.

3.3 Variable importance analysis

To improve model interpretability, permutation importance analysis was conducted to evaluate the contribution of each predictor to fire occurrence prediction (Hapfelmeier et al., 2023; Mehdiyev et al., 2025). Because the input variables consisted of both temporal and non-temporal spatial predictors, two permutation strategies were used. For temporal variables, including rainfall, air temperature, and soil moisture, the temporal order was randomized while the remaining variables were kept unchanged. For non-temporal spatial variables, including distance to human interaction and land-cover type, the spatial locations were randomized to evaluate the importance of their spatial structure.

The permutation importance results show that land cover type and rainfall were the most influential variables in the model. Randomizing land cover type produced the largest decrease in performance, with reductions of 0.085 in ROC–AUC. This indicates that spatial differences in land cover provide important information for distinguishing fire-prone and non-fire-prone areas. Among the temporal predictors, rainfall showed the strongest contribution, with decreases of 0.060 in ROC–AUC after temporal permutation. Distance to human interaction also had an important contribution, with decreases of 0.057 in ROC–AUC, indicating the relevance of anthropogenic proximity in explaining fire occurrence. Soil moisture and air temperature also contributed to model performance, although their effects were smaller.

3.4 Sensitivity analysis

We conduct sensitivity analysis for the next three-day (**Figure 13**) and the next ten-day (**Figure 14**) forecasts to assess the response of the predicted fire probability to controlled perturbations in the rainfall, air temperature, and soil moisture. Two contrasting scenarios are examined using perturbation levels of 10%, 30%, and 50%. These perturbations were applied relative to the original input values used for each forecast horizon. Therefore, the unperturbed next three-day and next ten-day forecast inputs were used as the reference scenarios, rather than historical mean values. The first scenario characterized by increased rainfall, increased soil moisture, and decreased air temperature, representing a wetter and cooler environmental state. In contrast, the second scenario represented by decreased rainfall, decreased soil moisture, and increased air temperature, representing a drier and warmer environmental state associated with lower water availability and enhanced evapotranspiration. These perturbations are designed to evaluate the model response under varying meteorological and hydrological conditions.

The sensitivity results demonstrate that the model exhibits a consistent response across both forecasting horizons, including the next three-day and the next ten-day. Under the first scenario, the predicted probability of fire decreases progressively with increasing perturbation, and the extent of susceptible areas contracts across the study region. Under the second scenario, the predicted probability of fire increases substantially, and the spatial extent of elevated fire risk areas expands with increasing perturbation magnitude from 10% to 50%.

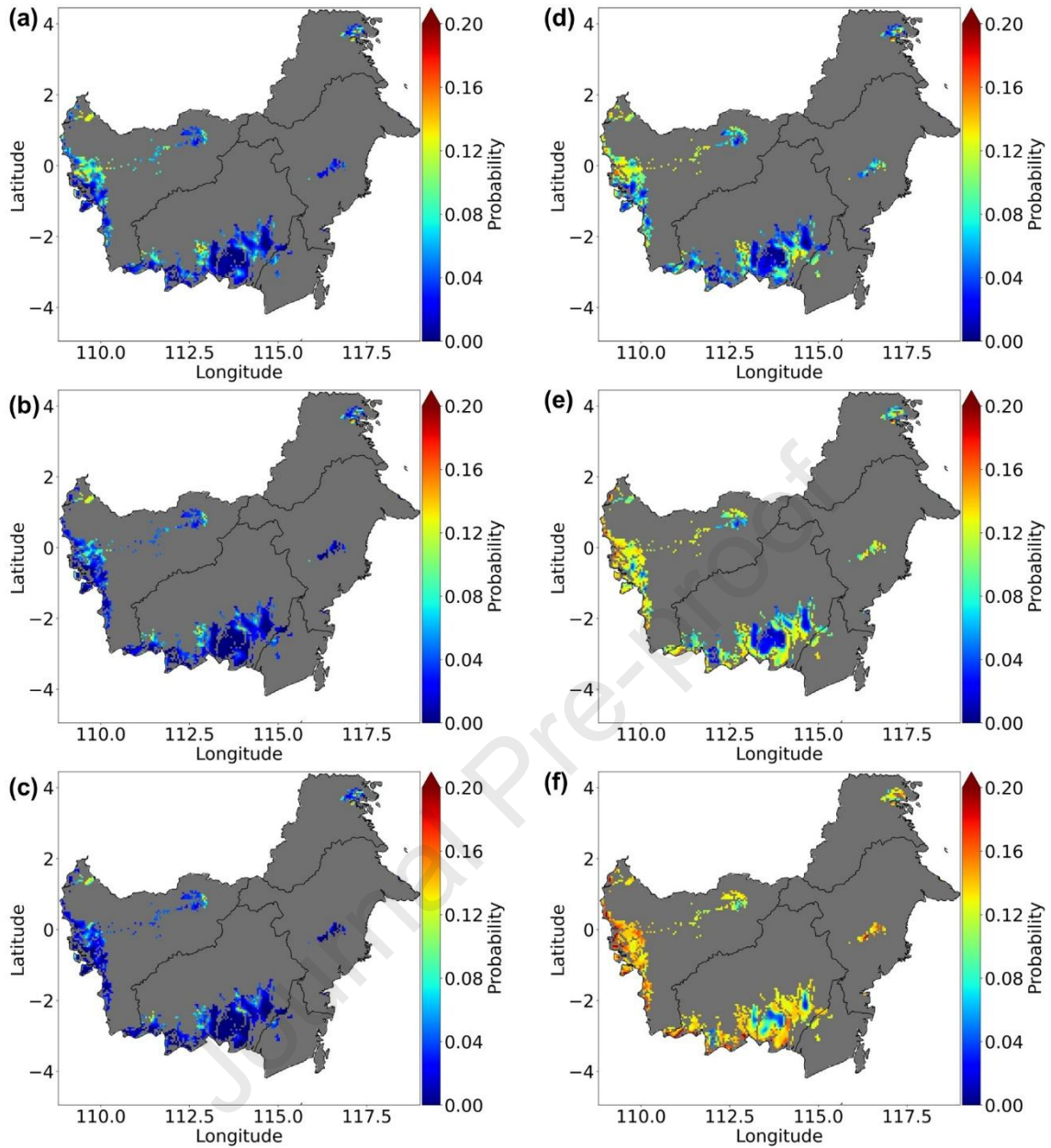


Figure 13. Sensitivity analysis for the next three-day forecast under the first scenario (a, b, c) and the second scenario (d, e, f) with perturbation levels of 10%, 30%, and 50%, respectively. The first scenario represents wetter and cooler conditions, with increased rainfall, increased soil moisture, and decreased air temperature. Contrastingly, the second scenario represents drier and warmer conditions, with decreased rainfall, decreased soil moisture, and increased air temperature.

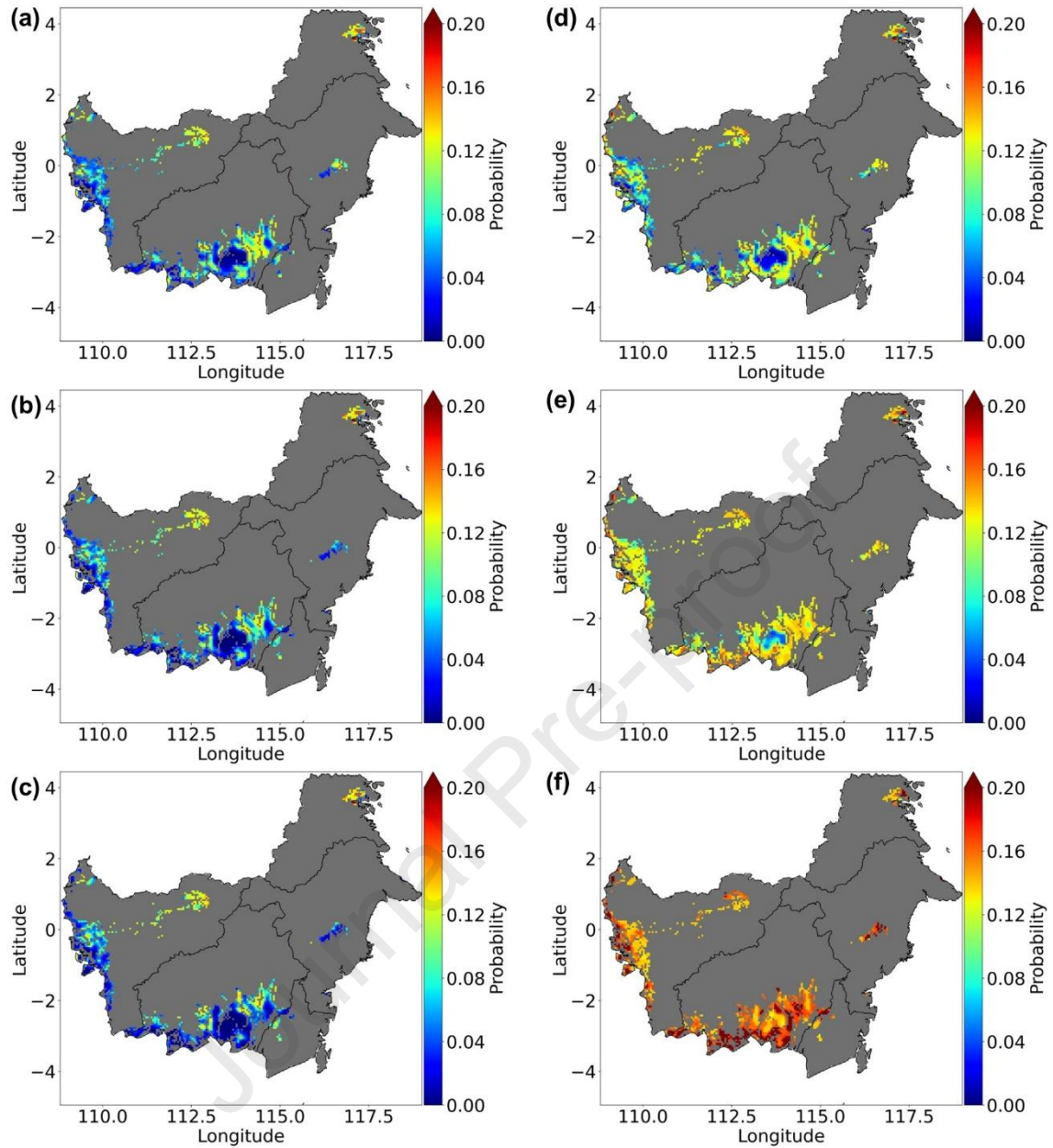


Figure 14. Sensitivity analysis for the next ten-day forecast under the first scenario (a, b, c) and the second scenario (d, e, f) with perturbation levels of 10%, 30%, and 50%, respectively. The first scenario represents wetter and cooler conditions, with increased rainfall, increased soil moisture, and decreased air temperature. Contrastingly, the second scenario represents drier and warmer conditions, with decreased rainfall, decreased soil moisture, and increased air temperature.

4. Discussion

The proposed model integrates Convolutional Neural Networks (CNN) (Khirani et al., 2026; Yar et al., 2023; Zelioli et al., 2025), Convolutional Long Short-Term Memory (ConvLSTM) (Khalaf et al., 2024; Mambile et al., 2025; Xu et al., 2025), and isotonic regression (DeLong & V. Wüthrich, 2025; Gneiting & Resin, 2023) to provide a unified representation of the spatial, temporal, and probabilistic characteristics of tropical peatland fire. The CNN component extracts spatially structured features from gridded environmental predictors, allowing the model to learn patterns associated with fire prone conditions. These spatial features are subsequently processed by the ConvLSTM component, which captures temporal evolution while preserving spatial dependence. Consequently, the model appropriately represents the development, persistence, and shift of fire susceptibility. The inclusion of isotonic regression further improves the framework by calibrating the predicted probabilities and increasing the consistency of the model outputs with the observed frequency of fire events.

In comparison with existing approaches for assessing tropical peatland fire risk, including the Peat Fire Vulnerability Index (PFVI) (Taufik et al., 2022) and machine learning models (Horton et al., 2021; Mahdiyasa et al., 2025), the proposed model provides a more comprehensive representation of fire dynamics (**Table 3**). PFVI provides simple and physically interpretable measures of fire danger under drying conditions, while machine learning models offer greater flexibility for modeling nonlinear relationships among variables. However, these approaches often capture only certain aspects of tropical peatland fire behaviour, either through index-based representations of vulnerability or through models that focus primarily on spatial or temporal patterns separately. The proposed model extends these capabilities by integrating spatial feature extraction, temporal dependence, and probability calibration within a single framework.

Table 3. Comparison of the proposed model, the spatial machine learning model of Horton et al. (2021) the temporal machine learning model of Mahdiyasa et al. (2025) and the Peat Fire Vulnerability Index (PFVI) of Taufik et al. (2022) for analyzing tropical peatland fire dynamics.

	Spatial dependence	Temporal dependence	Fire risk
Proposed model	Explicitly represented through CNN-based spatial feature extraction	Explicitly represented through ConvLSTM-based temporal learning	Calibrated probability of fire through isotonic regression
Spatial machine learning model from Horton et al. (2021)	Explicitly represented through XGBoost and Random Forest methods	Not incorporated in the model	Uncalibrated predicted probability
Temporal machine learning model from Mahdiyasa et al. (2025)	Not incorporated in the model	Explicitly represented through LSTM and GRU methods	Fire index or fire danger class
Peat Fire Vulnerability Index from Taufik et al. (2022)	Not incorporated in the model	Explicitly represented through empirical relationship	Fire index or fire danger class

579

580 In addition to empirical indices and machine learning models, the integration of IoT-
581 based monitoring systems and the Fire Weather Index (FWI) has also been widely discussed
582 for peatland fire monitoring and early warning. A recent study highlighted the integration of
583 IoT-based environmental monitoring and FWI systems for improving fire prediction and
584 mitigation (Li, 2025). Other studies have shown that IoT-enabled systems combined with
585 neural networks can predict groundwater level using atmospheric parameters with root mean
586 square error values between 3.554 and 4.920 mm, supporting real time fire risk mitigation in
587 peatlands (Kumari et al., 2025). Similarly, Li et al. (2023a) developed an IoT system in the
588 Raja Musa Forest Reserve, Malaysia, and proposed a low-complexity model to calculate the
589 Duff Moisture Code using groundwater level and relative humidity, achieving a correlation of
590 0.91 with actual values. Li et al. (2023b) further demonstrated that multi-layer neural networks
591 can predict FWI from IoT-measured fire risk factors, where a nine-input five-layer network
592 outperformed simpler models with an mean square error of 1.116 and a correlation of 0.890. In
593 addition, Li et al. (2022) showed that groundwater level can be incorporated into tropical
594 peatland FWI formulation by replacing the Drought Code inputs of temperature and rainfall
595 with groundwater level and validating the approach using LoRa-based IoT data and machine
596 learning prediction. These IoT- and index-driven approaches are useful for identifying local
597 drying conditions and supporting site specific early warning. However, their application over
598 extensive tropical peatland landscapes may be constrained by sensor deployment density,
599 maintenance requirements, network connectivity, and regional calibration. The proposed
600 framework complements these approaches by using spatially continuous gridded
601 meteorological, hydrological, land cover, anthropogenic, and historical fire data to learn large
602 area spatio-temporal fire risk patterns without requiring dense in-situ sensor networks.

This advantage is reflected in the resulting probability of fire maps (**Figure 9**), which represent fire likelihood across the study area. The probability based output provides a more informative representation compared to categorical fire danger levels (Mahdiyasa et al., 2025; Mashuri et al., 2026; Taufik et al., 2022), because it indicates not only areas with elevated fire risk but also the relative likelihood of occurrence across the landscape. This allows the proposed model to distinguish areas with different levels of susceptibility and to represent gradual variation in fire risk. Furthermore, the agreement between predicted fire prone areas and the observed hotspot distribution (**Figure 11**) suggests that the proposed model is able to capture the dominant patterns of tropical peatland fire occurrence and translate them into reliable probability estimates.

4.1 Forecasting and early detection of tropical peatland fire

The forecasting results (**Figure 11**) indicate that the location and extent of susceptible areas vary across prediction horizons, suggesting that the model captures the evolution of fire prone conditions under changing environmental conditions. This behaviour is important in tropical peatlands, where fire susceptibility is influenced by short-term drying, prolonged moisture deficits, and seasonal variability (Harrison et al., 2024; Mezbahuddin et al., 2023; Ramdzan et al., 2023). Under these conditions, areas that initially exhibit low probability of fire may become increasingly vulnerable over time, while other areas may indicate declining susceptibility. Consequently, the model represents fire risk as a dynamic process that changes with hydrological and meteorological conditions. This capability increases the practical relevance of the model for early warning by capturing both current fire risk and its probability evolution over time.

From an operational perspective, the forecasting capability of the proposed model is valuable for early detection and targeted fire management in tropical peatland regions (Horton

et al., 2022; S. Mishra et al., 2021). By anticipating the expansion or contraction of susceptible zones over time, the model provides actionable information for monitoring and mitigation processes. Areas projected to experience decreasing susceptibility may be assigned lower priority for operational response. In contrast, areas with persistently high fire probability may be monitored as critical zones requiring continuous attention. As a result, the model not only produces fire risk maps but also supports a more adaptive approach to peatland fire management.

4.2 Sensitivity of tropical peatland fire probability maps to different climate conditions

Wetter and cooler climate conditions reduce tropical peatland fire risk by enhancing moisture retention within the peat pore space (Horton et al., 2022; Taufik et al., 2022). Higher rainfall, increased soil moisture, and lower air temperature contribute to wetter peat conditions, reducing the likelihood of ignition and limiting fire spread across peatland areas. Under these conditions, peat is less vulnerable to sustained burning, and the extent of areas with elevated fire susceptibility could decrease. These effects can reduce both the probability of fire occurrence and the spatial persistence of high-risk zones across the landscape. A corresponding spatial pattern is the expansion of areas classified as having zero probability of fire occurrence, with proportions increasing from about 22% to 42% for the next three-day forecast (**Figure 13a, b, c**) and from about 18% to 35% for the next ten-day forecast (**Figure 14a, b, c**) across perturbation levels from 10% to 50%. Consequently, a larger portion of the peatland landscape becomes effectively resilient to fire under wetter and cooler conditions.

In contrast, drier and warmer climate conditions increase tropical peatland fire risk by intensifying peat drying and reducing moisture availability (Kuwata, 2024; Mezbahuddin et al., 2023; Schmidt et al., 2024). A combination of decreased rainfall, lower soil moisture, and higher air temperature accelerate the loss of water from the peat surface and enlarge the

unsaturated zone, creating conditions that are more favourable for ignition and fire spread. Under these conditions, peat becomes more vulnerable to persistent burning, particularly during periods of prolonged moisture deficit. In addition to increasing the probability of fire occurrence, these effects also enlarge the extent and persistence of susceptible areas across the landscape. A notable pattern is the substantial reduction in areas classified as having zero probability of fire occurrence, with proportions decreasing from about 10% to 1% for the next three-day forecast (**Figure 13d, e, f**) and from about 8% to 0% for the next ten-day forecast (**Figure 14d, e, f**) over perturbation levels ranging from 10% to 50%. This pattern indicates that regions previously considered effectively free from fire risk become increasingly susceptible under drier and warmer conditions. As a result, drier conditions could lead to more widespread and severe fire prone conditions in tropical peatlands.

4.3 Limitations and future research

The application of isotonic regression for calibration purposes depends on the availability and representativeness of the calibration data. In highly imbalanced tropical peatland fire datasets, where fire events are relatively rare, isotonic regression may be sensitive to sparse observations (Tian & Zhang, 2022; Zhang et al., 2025). This condition could produce stepwise calibration functions that are unstable and may not be consistent under different spatial or temporal conditions. Future research could develop calibration models that vary across regions, seasons, or peatland conditions (Little et al., 2024; Tohme et al., 2020). This would allow the calibrated probabilities to more accurately represent fire risk characteristics and changing environmental conditions, which improves the reliability of probability of fire maps for operational early warning systems.

Although the proposed model can learn spatial and temporal patterns from meteorological, hydrological, land cover, anthropogenic, and fire data, it may not fully capture

peatland structural changes, including variations in bulk density, active porosity, and hydraulic conductivity. These properties are important because they affect water retention, groundwater dynamics, peat drying, and the susceptibility of peat to fire (Baird et al., 2017; Crawford et al., 2024; Rezanezhad et al., 2016; Wijedasa, 2016). Future research could integrate the proposed data-driven framework with physics-based peatland models to provide a more process-based representation of tropical peatland fire risk (Kurnianto et al., 2015; Mahdiyasa et al., 2022, 2023, 2024; Young et al., 2023). Furthermore, the framework should also incorporate land use dynamics, critical variables in peatland conversion that are closely linked to increased peat fire risks. This integration would allow the model to account not only for observed fire patterns but also for the ecohydrological and mechanical processes that regulate peatland condition. By incorporating changes in peat physical properties and hydrological behavior, future models could improve the interpretation of fire risk dynamics and support more reliable long-term forecasting under different environmental and land management scenarios.

The spatial resolution used in this study may influence the representation of small fire patches. All input datasets were resampled to a uniform 5×5 km grid to ensure spatial consistency among data sources with different native resolutions. Similar grid sizes have been used in previous fire related studies (Bar et al., 2023; Tan et al., 2020; Wickramasinghe et al., 2024), indicating that the selected resolution is suitable for regional scale fire risk mapping and for integrating meteorological, hydrological, land cover, anthropogenic, and fire occurrence variables within a common modeling framework. However, the 5 km grid should be interpreted as a grid cell level fire occurrence probability unit rather than a fine scale fire patch detection resolution. In this study, a grid cell was labeled as positive when it contained at least one hotspot detection during the corresponding time step. Consequently, the model estimates the probability of fire occurrence within each grid cell rather than delineating the exact size, shape, or boundary of individual fire patches. Small or isolated fire events may be aggregated with

surrounding non-fire conditions, which can smooth fine scale spatial variability and reduce the ability of the model to represent localized fire clusters. Further research should examine the sensitivity of model performance to alternative grid resolutions and evaluate their effects on hotspot.

Future could also develop the proposed model into an operational dashboard for a tropical peatland fire early warning system. This development would allow users to monitor fire risk conditions dynamically, identify areas with increasing fire probability, and support timely prevention or mitigation actions (Horton et al., 2022). The dashboard might include alert levels based on calibrated fire probabilities, enabling the model outputs to be translated into practical decision support information for land managers, government agencies, and plantation operators (Purnomo et al., 2024). Therefore, the development should focus on real time data integration and user-friendly visualization to ensure that the dashboard is reliable for operational tropical peatland fire management.

5. Conclusion

This study developed a spatio-temporal framework to analyze and predict tropical peatland fire risk by integrating a Convolutional Neural Network (CNN) for spatial feature extraction, Convolutional Long Short-Term Memory (ConvLSTM) for temporal dependency learning, and isotonic regression for probability calibration. Through this integrated structure, the proposed model is able to identify areas with increased fire probability and produces probability of fire maps that are consistent with observed hotspot distributions. Furthermore, the probability calibration method employed by this framework improve the agreement between model outputs and observed fire event frequencies, which support risk assessment in tropical peatland ecosystems. Therefore, the proposed model provides not only a predictive

tool but also decision support information for improving tropical peatland fire mitigation and restoration management.

CRedit authorship contribution statement

Adilan Mahdiyasa: Conceptualization, Methodology, Software, Formal analysis, Writing-original draft, Writing-review & editing.

Jeremy Chaim: Software, Methodology, Formal analysis, Writing-review & editing.

Udjianna Pasaribu: Methodology, Writing-review & editing.

Elham Sumarga: Methodology, Writing-review & editing.

Bagus Muljadi: Methodology, Resources, Writing-review & editing.

Asrini Chrysanti: Methodology, Writing-review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Peatland distribution data are obtained from Geospatial Information Agency of Indonesia (BIG) at <https://geoportal.big.go.id/#/>. Fire hotspots data are obtained from Fire Information for Resource Management System (FIRMS) at <https://firms.modaps.eosdis.nasa.gov/>. Rainfall data are obtained from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) at <https://www.chc.ucsb.edu/data/chirps>. Air temperature and soil moisture data are obtained from ERA5-Land reanalysis dataset at <https://cds.climate.copernicus.eu/>. Land cover

classification data are obtained from MODIS Collection 6 Land Cover Type Product at <https://modis.gsfc.nasa.gov/data/dataproduct/mod12.php>. Distance to human activity data are obtained from Global Human Settlement Layer (GHSL) at <https://human-settlement.emergency.copernicus.eu/>.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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